

Open Source Tools in GIS

Here's an overview of the open source GIS tools for data visualisation, storage and file conversion.

Do you use a GPS navigator on your phone? Do you use Google Maps to plan a weekend trip? Do you find yourself toying with Google Earth and taking virtual flights?

If you do, then you are already using a Geographic Information Systems (GIS) tool. A GIS would have most of the following features: data acquisition, data storage, data conversion, analysis functions, visualisation and image processing. GIS can serve as a Decision Support System (DSS) in applications ranging from tour planning and city planning, to establishing a worldwide environment policy.

GIS is fascinating as it is data-rich, uses colour-coded visualisation and, most importantly, it represents an aspect of the real world we live in. So let's get familiar with some of the open source options in data visualisation/editing, spatial databases and spatial file conversion.

Spatial data is the core of any GIS. It could be available in a raster format like GeoTiff, JPEG2000 and DRG, or a vector format like GML, KML, Shapefile, TIGER, GeoJSON, etc.

GRASS (Geographic Resources Analysis Support System) is the leading and most comprehensive open source GIS. It is vast and complex, and is beyond the scope of this article, so I will not cover it here.

Data visualisation and editing

A GIS visualisation and editing tool will allow you to read one or more spatial file formats and let you query and manipulate the

data. You can even generate maps with custom colour schemes and labels.

uDig

uDig is a desktop-based GIS built on the Eclipse RCP. It allows you to connect to a WMS (Web Map Service), WFS (Web Feature Service), WCS (Web Coverage Service), spatial database, GeoRSS or Shapefile. It's best used to generate maps with beautiful colour schemes and to draw new map layers. It's very easy to install and use.

uDig does not have analysis functions of its own but has a number of plug-ins to beef it up. JGrass interfaces readily with uDig. The JGrass plug-in uses the GRASS algorithms and the uDig user interface to provide hydrological and geomorphological analysis.

uDig supports Common Query Language (CQL) to query spatial vector data.

QGIS

Quantum GIS (QGIS) is another desktop-based GIS that can read most widely used spatial file formats, and allows you to create and edit vector and raster data. QGIS lets you query the attribute table of vector files using the SQL syntax and the result can be visualised or saved as a new spatial data file. QGIS, with the help of the gpsable tool, allows you to import and view your GPS trails; furthermore, you can generate your own travel maps.

QGIS comes with Python binding and thus provides a Python console for scripting, and has a long list of handy plug-ins. QGIS integrates well